

**DATA VORTEX A'26
ROUND 2
REBUILDING THE SOCIAL ENGINE**

EVALUATION METRICS REPORT

Generated from validated local evaluation run
2026-09-18 UTC

Dataset summary

The supplied labelled CSV has 9,000 rows and 4 columns. The evaluated supervised labels are sentiment_label, topic_category. Normalized duplicate text groups were kept intact in group-aware splits; no duplicate group crosses a split.

Evaluation methodology

Models were selected on a validation partition using macro F1. The held-out test partition was not used in selection. Accuracy, macro precision, macro recall, macro F1, weighted F1, and per-class scores were then calculated directly from final test predictions.

Data split

Partition	Rows
train	6316
validation	890
test	1794

sentiment_label: validation comparison

Model	Accuracy	Macro P	Macro R	Macro F1	Weighted F1	Train s
TF-IDF + Logistic Regression	0.5809	0.5781	0.5759	0.5763	0.5833	0.88
TF-IDF + Linear SVM	0.5708	0.5696	0.5662	0.5670	0.5736	0.48
Majority baseline	0.2876	0.0959	0.3333	0.1489	0.1285	0.33

Final model and held-out test metrics: sentiment_label

Final model: TF-IDF + Logistic Regression. Choice was frozen before the final test evaluation.

Metric	Value
Accuracy	0.5786
Macro precision	0.5817
Macro recall	0.5783
Macro F1	0.5795
Weighted F1	0.5800

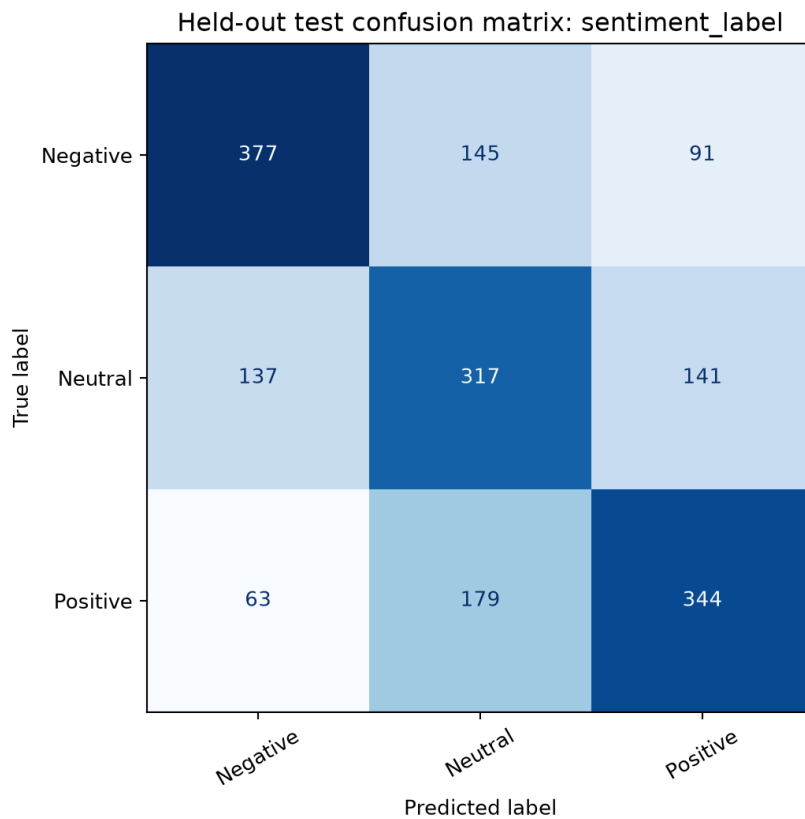


Figure: confusion matrix generated directly from held-out test labels and predictions.

Per-class metrics

Class	Precision	Recall	F1	Support
Negative	0.6534	0.6150	0.6336	613
Neutral	0.4945	0.5328	0.5129	595
Positive	0.5972	0.5870	0.5921	586

Error-analysis summary

There were 756 misclassified held-out posts. Most frequent observed confusion pairs: Positive → Neutral: 179; Negative → Neutral: 145; Neutral → Positive: 141. The structured, non-cherry-picked error table is saved in results/error_analysis_sentiment_label.csv (up to 25 highest-margin errors) and all test predictions are in results/predictions_sentiment_label.csv.

topic_category: validation comparison

Model	Accuracy	Macro P	Macro R	Macro F1	Weighted F1	Train s
TF-IDF + Linear SVM	0.9146	0.8530	0.5128	0.5970	0.8951	6.85
TF-IDF + Logistic Regression	0.9011	0.7336	0.5129	0.5842	0.8872	10.77
Majority baseline	0.8494	0.2124	0.2500	0.2296	0.7803	3.56

Final model and held-out test metrics: topic_category

Final model: TF-IDF + Linear SVM. Choice was frozen before the final test evaluation.

Metric	Value
Accuracy	0.9264

Metric	Value
Macro precision	0.9147
Macro recall	0.4919
Macro F1	0.5805
Weighted F1	0.9110

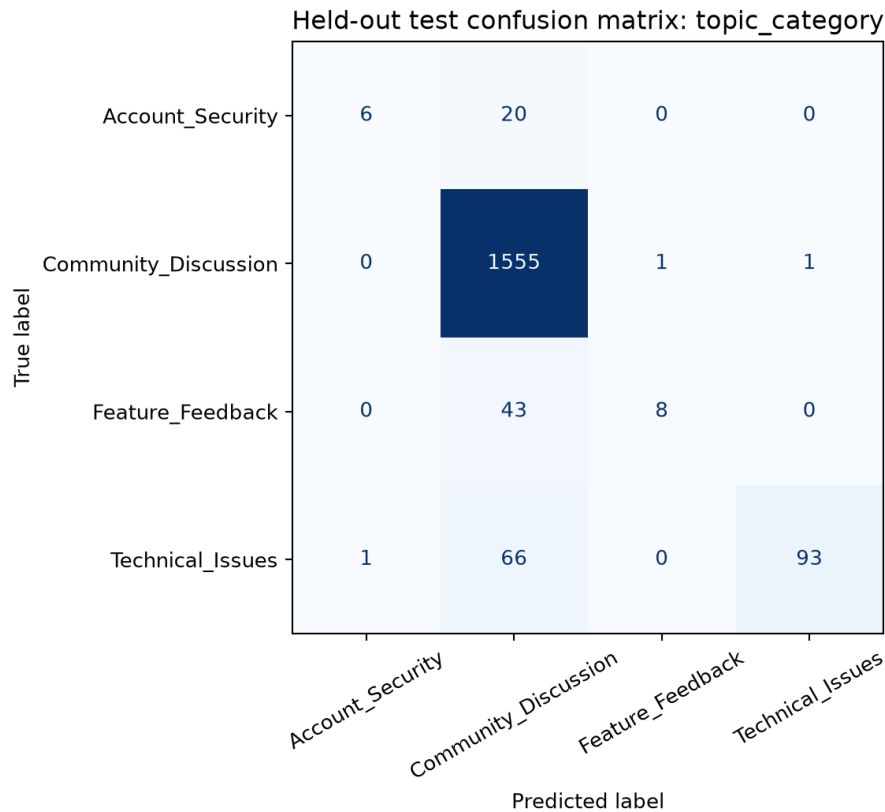


Figure: confusion matrix generated directly from held-out test labels and predictions.

Per-class metrics

Class	Precision	Recall	F1	Support
Account_Security	0.8571	0.2308	0.3636	26
Community_Discussion	0.9234	0.9987	0.9596	1557
Feature_Feedback	0.8889	0.1569	0.2667	51
Technical_Issues	0.9894	0.5813	0.7323	160

Error-analysis summary

There were 132 misclassified held-out posts. Most frequent observed confusion pairs: Technical_Issues → Community_Discussion: 66; Feature_Feedback → Community_Discussion: 43; Account_Security → Community_Discussion: 20. The structured, non-cherry-picked error table is saved in results/error_analysis_topic_category.csv (up to 25 highest-margin errors) and all test predictions are in results/predictions_topic_category.csv.

Key findings

Both targets are reported as separate supervised NLP tasks. Macro metrics are emphasized because they weight every label equally and make class-level weaknesses visible.